8. Compute test statistic χ^2

$$\chi^2 = \sum \frac{(\text{number of observations} - ef)^2}{ef}$$

$$\chi^2 = \frac{(a - ef_{11})^2}{ef_{11}} + \frac{(b - ef_{12})^2}{ef_{12}} + \frac{(c - ef_{21})^2}{ef_{21}} + \frac{(d - ef_{22})^2}{ef_{22}}$$

David M. Lane (Rice University) provided an alternative calculation of the test statistic χ^2 that makes a correction for continuity. This calculation is:

$$\chi^2 = \frac{n\left(|ad - bc| - \frac{n}{2}\right)^2}{(a + b)(c + d)(a + c)(b + d)}$$

Comparison to Critical Value:

- df = (number of columns − 1) · (number of rows − 1) = 1.
- Reject the null hypothesis if $\chi^2 \geq \chi^{2*}(\text{df} = 1,\ \alpha = 0.05) = 3.84$.

Distribution Analysis

Distribution analysis compares the entire set of performance data by determining if the set of outcomes could have been generated from the same data generating process ("DGP"). These tests could be based on either pair-samples or independent samples. Analysts need to categorize results based on price trends, capitalization, side, etc. in order to determine if one set of algorithms performs better or worse for certain market conditions or situations. Many times a grouping of results may not uncover any difference in process.

Chi-Square Goodness of Fit

The chi-square goodness of fit test is used to determine whether two data series could have been generated from the same underlying distributions. It utilizes the probability distribution function (pdf). If it is found that the observations could not have been generated from the same underlying distribution then we conclude that the algorithms are different.

Hypothesis:

H_0: Data generated from same distribution
H_1: Data generated from different distributions